

Manual

Controlling of Machine Data MDE-CMD 8.1

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1 Overview of Machine Data Controlling

Purpose

Machine data controlling provides the ability to evaluate data that was entered in the system as concerns how recorded durations are distributed to various time accounts. It also analyzes any statuses that occurred with respect to the duration and number of the occurrences.

Implementation considerations

You use the function package if:

- You would like to perform a downtime / status assessment of separate machines or machine groups over a choice of time periods.
- You would like to perform a status class evaluation for a selected machine or group of machines over a choice of time periods or shifts.
- You would like to perform an evaluation of the time accounts of a selected machine or group of machines over a choice of time periods or shifts.

Integration

Times and quantities entered and statuses posted in MDE are accessed for presentation.

Features

- Status report (workplace/machine-related)
 - Tabular downtime / status assessment of separate machines over a choice of time periods. Graphical comparison of production times and downtimes. Graphical presentation of a machine hit list that shows the machines with either the longest or the most downtimes.
- Status report
 - Tabular downtime / status evaluation of a selected machine or group of machines across a choice of time periods or shifts. Graphical comparison of production times and downtimes. Graphical presentation of a status hit list that shows the longest or the most downtimes.
- Status class report
 - Tabular and graphical status class evaluation of a selected machine or group of machines over a choice of time periods or shifts.
- RPA report
 - Tabular and graphical RPA evaluation of a selected machine or group of machines over a choice of time periods or shifts.

- Status profile
 - Chronological evaluations of status / downtimes of separate machines or machine groups. Staging of data in table and pivot table form. Data evaluation in pivot table based on various criteria. Graphical data presentation in pivot table form.
- Status class profile
 - Chronological evaluations of status classes for separate machines or machine groups. Staging of data in table and pivot table form. Data evaluation in pivot table based on various criteria. Graphical presentation of data in pivot table form.
- RPA profile
 - Evaluation of resource performance accounts of separate machines or machine groups in tabular and graphical form.
- Status analysis
 - Evaluation of status / downtimes using a pivot table. Definition of the pivot table presentation based on various criteria. Graphical presentation (bar chart).
- Status hit list
 - Table showing cumulative times over a time period for active machine statuses, each sorted by duration and restricted to the x longest durations.
- Minor/major stops
 - Table showing cumulative times over a time period for active workplace / machine statuses, broken down into short (minor) and long (major) statuses.
- ABC analysis
 - Table showing cumulative times over a time period for active workplace / machine statuses, categorized into three ABC classes, each depending on the relationship of the durations in percent. Drill-down option used to list separate status times in a detail table.

2 Status Report, Machine-Related

Summary

Menu	Operating facilities management → Status analyses → Status report (machine-related)
Transaction code	mstrpm
Function authorization	mstrpm

Usage

The status report (machine-related) is an evaluation in operating facilities management. The analysis provides summarized status information relating to a workplace/ machine for a certain period of time and a certain number of workplaces. The data is presented in charts for informational purposes. You can also compare several machines with each other using this status report.

Selection criteria

The application provides the following selection criteria:

Workplace

Search by machine/ by workplace. You can also run a search using wildcards.

Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can also run a search using wildcards.

Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can also run a search using wildcards.

Company

Search by workplaces/ machines that are assigned to the company that was entered. You can also run a search using wildcards.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Short designation

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

Designation

This field references the short name of the machines and workplaces in the machine's master data. At the bottom, only those machines are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders *) in this field.

Inclusive status for RPA 11

Accounts for the status for RPA 11 (usually, this is the "Production" status).

Date

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, while when selecting by time the selection is based on the start date. Please keep in mind that a selection by shift is only supported for ADE and MDE data, not for WRM data.

Shift(s)/ time

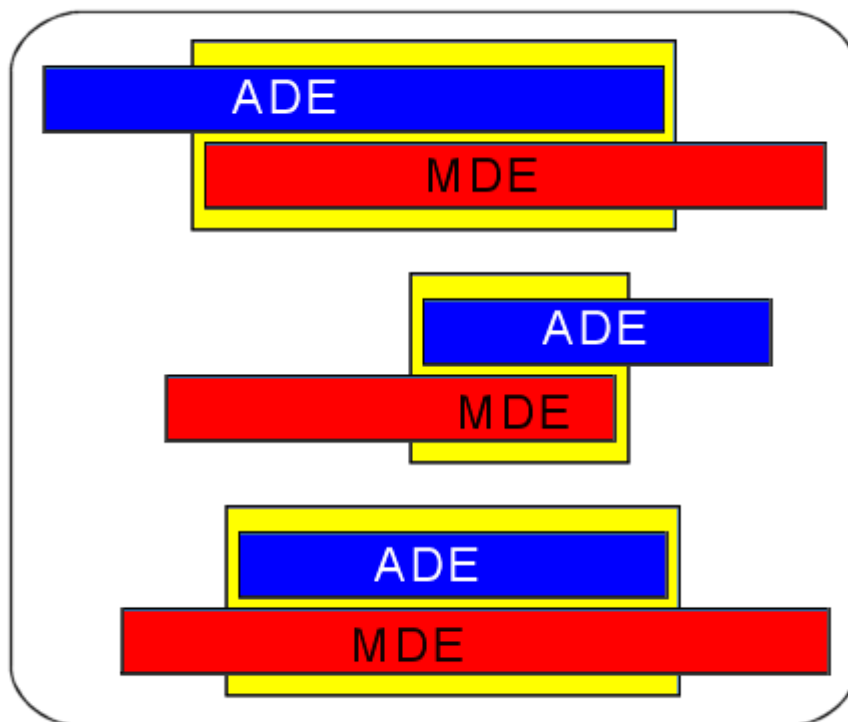
Selection by shift (only ADE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

Order/ article

For this kind of evaluation type, only the finished ADE postings are considered. If the order is currently still running on the machine, the time period between the last logon and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only ADE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the ADE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.

The following illustration shows an example of an overlapping of ADE and MDE postings. The ADE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

Resource type/ resource

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

Additional selection notes

Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Determining a shift-adjusted quantity

This option known from MDE 7.2 console is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.

Status report - machine related detail application

In the tabular evaluation, the detailed status of a workplace/ a machine or group is presented within an arbitrarily selected period of time. The durations and quantities (frequencies) are compressed to each separate status.

The following data is available:

Machine/ workplace category

Workplace

Number of the machine/ workplace.

Group

Group that the workplace is assigned to based on the resource configuration.

Cost center

Cost center that the workplace is assigned to based on the resource configuration.

Company

Company that the workplace is assigned to based on the resource configuration.

Duration category

Duration

Total time of all disturbances within the evaluation period (i.e. not all downtimes within this evaluation period are compressed in this field).

%

Proportion of total downtime durations for the specific machine in relation to the total duration of disturbances of all of the displayed workplaces shown in percent.

Production

Durations posted to RPA 11

RPA

When delivered from the factory, the durations set for the resource performance accounts 1-10 are listed in the pool of columns. They are stored there with their abbreviation and can be shown as needed.

Please keep in mind that there are resource performance account columns for duration and quantities.

Quantity category

Quantity

Total number of disturbances that was active at each machine within the evaluation period.

%

Proportion of the number of downtimes for the specific machine in relation to the total number of all of the workplaces displayed shown in percent.

Production

Number of RPA 11 statuses in the evaluation period.

RPA

When delivered from the factory, the number of resource performance accounts 1-10 is listed in the pool of columns. They are stored there with their abbreviation and can be shown as needed.

Please keep in mind that there are resource performance account columns for duration and quantities.

Rate of capacity utilization

Rate of capacity utilization in %

Comparison production - downtimes detail application

The "Comparison: production - downtimes" detail application provides a summary of durations for production (RPA 11) and for the downtimes (RPA 1-10) for the workplaces marked in the table and these are compared graphically in the form of a bar chart.

The percentages shown relate to the total of both bars.

Machine hit list (durations) detail application

The downtime durations for each of the workplaces marked in the table are shown in the form of a bar chart in the machine hit list (durations) detail application.

In the combo box "Displayed series" you can define whether the durations displayed as a percentage should relate to the total number of downtimes (RPA 1-10) or to the total duration (RPA 1-11). If both options have been selected, both bars are displayed (for each workplace).

Machine hit list (quantity) detail application

In the machine hit list (quantity) detail application, the number of downtimes per machine is displayed in a chart. As opposed to the previous detail applications, no further display options are available here.

3 Status Report

Summary

Menu	Operating facilities management → Status analyses → Status report
Transaction code	mstrp
Function authorization	mstrp

The status report is an indispensable tool for any production executive. Whether planner, foreman or team leader, this report can be individually configured to show downtimes in any form.

Usage

This evaluation provides status information relating to a workplace/ machine for a specific period of time and a certain number of workplaces. All of the accrued downtimes are pulled together in this evaluation and can be prepared in the form of a graphic or in table form based on what the user intends to achieve.

Selection criteria

The application provides the following selection criteria:

Workplace

Search by machine/ by workplace. You can run a search using wildcards.

Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can run a search using wildcards.

Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can run a search using wildcards.

Company

Search by workplaces/ machines that are assigned to the company that was entered. You can run a search using wildcards.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Short designation

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

Designation

This selection criterion references the name of the machines and workplaces in the machine's master data. At the bottom, only those machines are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders *) in this field.

Status

Limits the selection to a certain status. Displayed in the combo box are the status texts that the selection is filtered through.

Include status for RPA 11

Accounts for the status for RPA 11 (usually, this is the "Production" status).

Date

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, whereas when selecting by time, the selection is based on the start date. Please keep in mind that a selection by shift is only supported for ADE and MDE data, not for WRM data.

Shift/ time

Selection by shift (only ADE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

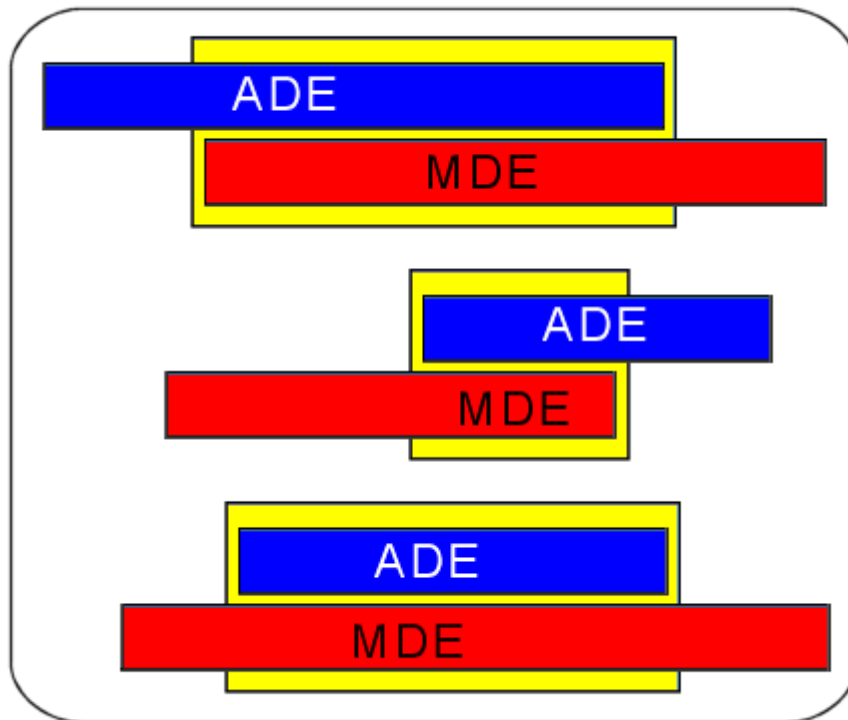
Order/ article

When you choose the option "Order", you must enter an order or an article.



For this kind of evaluation type, only the finished ADE postings are considered. If the order is currently still running on the machine, the time period between the last logon and "now" is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only ADE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the ADE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.

The following illustration shows an example of an overlapping of ADE and MDE postings. The ADE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

Additional selection notes

Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well.

Determining a shift-adjusted quantity

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.

The status assigned to RPA 12

Status assigned to RPA 12 (typically the status for breaks and for shift-free times) are not selected.

Status report detail application

Tabular evaluation of the downtimes for a specific period of time and a certain number of workplaces. The result depends on the selection and therefore on the selection criteria made available on the selection panel.

The following data are available:

Status category**Status**

Status number as per configuration

Designation

Designation of the status

RPA

Resource performance account number

Status class

Abbreviation of the status class

Duration category**Duration**

Total time of all statuses determined within the evaluation period (i.e. not all downtimes within this evaluation period are compressed in this field).

%

Proportion of total status duration as compared to the total duration of disturbances shown as a percentage.

Quantity category**Quantity**

Total number of statuses determined that were applicable during the evaluation period.

%

Proportion of the number of statuses to the total number shown as a percentage.

Comparison production - downtimes detail application

In this detail application, the production time (green) accrued during the evaluation period and the sum total of all determined status times, not including RPA 11 and 12 (red) are compared in the form of a bar chart.

The presentation always relates to all of the displayed statuses (there is no highlighting option in this table).

Status hit list (durations) detail application

The downtime durations for each status marked in the table are shown in the form of a graphic in the status hit list (durations) detail application.

In the combo box "Displayed series" you can define whether the durations displayed as a percentage should relate to the total number of downtimes (RPA 1-10) or to the total duration (RPA 1-11). If both options have been selected, both bars are displayed (for each status).

Status hit list (quantity) detail application

The total number of status changes for each status marked in the table is shown in the form of a graphic in the status hit list (quantities) detail application.

As opposed to the previous detail applications, no further display options are available here.

4 Status Profile

Summary

Menu	Operating facilities management → Status analyses → Status profile
Transaction code	mstpf
Function authorization	mstpf

Usage

The status profile evaluates the statuses by shift, by day, by calendar week or by month over a certain period of time. The result depends on the selection and therefore on the selection criteria made available on the selection panel.

Selection criteria

The application provides the following selection criteria:

Workplace

Search by machine/ by workplace. There is an option to search using wild cards.

Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field group in the workplace/ machine configuration. You can run a search using wildcards.

Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can run a search using wildcards.

Company

Search by workplaces/ machines that are assigned to the company that was entered. You can run a search using wildcards.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Short designation

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

Designation

This selection criterion references the name of the machines and workplaces in the machine's master data. At the bottom, only those machines are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders *) in this field.

Status text

Limits the selection to a certain status. Displayed in the combo box are the status texts that the selection can be filtered through.

Inclusive status for RPA 11

Accounts for the status for RPA 11 (usually, this is the "Production" status).

Date

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, whereas when selecting by time, the selection is based on the start date. Please keep in mind that a selection by shift is only supported for ADE and MDE data, not for WRM data.

Shift(s)/ time

Selection by shift (only ADE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

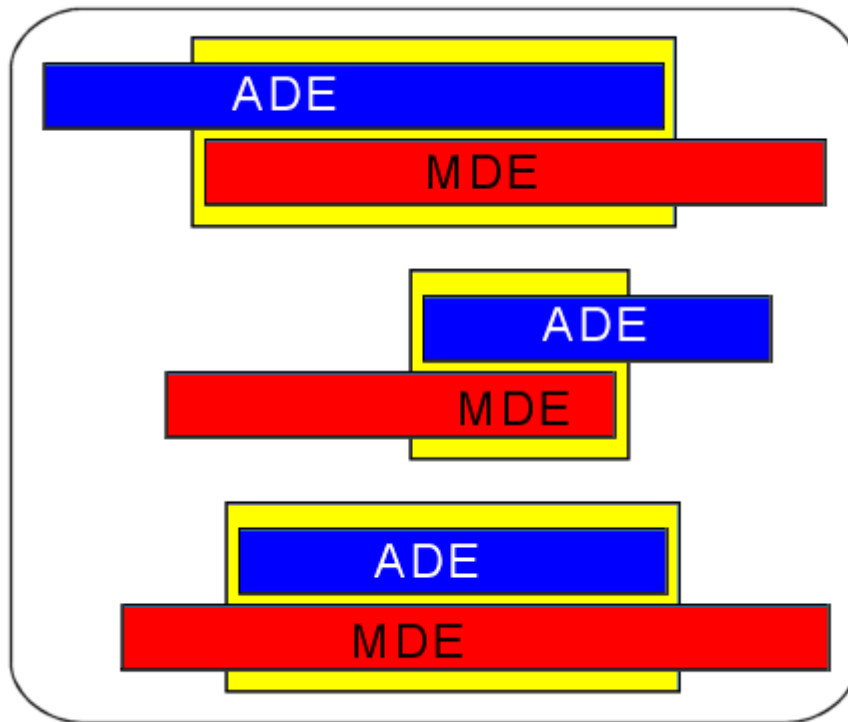
Order/ article

When you choose the option "Order", you must enter an order or an article.

For this kind of evaluation type, only the finished ADE postings are considered. If the order is currently still running on the machine, the time period between the last logon and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only ADE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the ADE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.



The following illustration shows an example of an overlapping of ADE and MDE postings. The ADE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type the resource postings take priority. The principle is the same as when running an evaluation by order.

Additional selection notes

Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Determining a shift-adjusted quantity

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.



Status profile detail application

Tabular presentation of the status with the following columns:

Shift date/ shift/ calendar week/ year

The durations in the statuses are displayed in groups by shift (i.e. shift date/ shift number). Calendar week, month and year (based on the shift date) are provided as additional sorting/ grouping criteria.

Status/ designation

Number and designation of the status. The status column is displayed in color as defined in the status text configuration.

RPA

Resource performance account number

Abbreviation

Abbreviation of the status class

Duration

Total time of all statuses within the evaluation period.

Quantity

Total number of statuses that was applicable during the evaluation period.

Pivot table detail application

You can evaluate status based on additional criteria in the pivot table detail view.



The bar colors in the chart are set "arbitrarily" using a color chart defined internally.

5 Status Hit List

Summary

Menu	Operating facilities management → Status analyses → Status hit list
Transaction code	sthitl
Function authorization	sthitl

Usage

The status hit list provides a quick overview of which statuses occurred most often at the machines and which state lasted the longest. It supplies information about the duration and the number of machine events recorded as a status. Also included in this overview are the production status (the status assigned to RPA 11) and the break status (the status assigned to RPA 12).

Statuses are both the machine/ workplace status, which is often also referred to as "Downtime reasons" or "Disturbances", but also other parallel statuses, such as program, operation type, operation mode or disturbance first value posting (depends on license/ project).

Selection criteria

The application provides the following selection criteria:

Workplace

Workplaces/ machines that match the criteria entered.

Group

Search by workplaces/ machines that are assigned to the group that was entered.

Date

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, whereas when selecting by time, the selection is based on the start date. Please keep in mind that a selection by shift is only supported for ADE and MDE data, not for WRM data.

Shift/ time

Selection by shift (only ADE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Top

Limits the number of statuses displayed for each selected machine to those with the longest duration. Pre-assignment: 5

Status type

Selection of status types that are included in the evaluation. By default, the machine status will be available here; additional status types are available depending on the license.

Field descriptions**Resource**

Workplace/ machine number

Resource Type

For workplaces/ machines, generally "MNR".

Designation

The name used to designate the workplace/ machine.

Status, status text

Status number and status text for the status that applied. The status is displayed in the status text color that was configured.

Duration

Length of time that the status applied.

Total number

Number of times that the status applied.

Status type

Description of the status type that the status belongs to. Typically this is "Machine status"; other status types are available depending on the license or the specific project.

6 ABC Analysis

Summary

Menu	Operating facilities management → Status analyses → ABC analysis
Transaction code	stabc
Function authorization	stabc

Usage

All statuses are listed in this evaluation that occur while a selected machine is running. The statuses are broken down based on the “Pareto principle” – which means they are sorted, added up and classified by their size and divided into the three categories A, B and C by the duration of their occurrence. Threshold values may be selected.

Selection criteria

The application provides the following selection criteria:

Workplace

Defines for which workplace the ABC analysis is shown.

Status type

Narrows down the error messages displayed to a status type (depends on license or project).

Threshold value 1

Parameter used to set the ABC thresholds. For threshold 1, the threshold is defined between the limits A and B. The predefined value is 50%.

Threshold value 2

Parameter used to set the ABC thresholds. For threshold 2, the threshold is defined between the limits B and C. The predefined value is 30%.

Date from - to/ shift/ time

Narrows down the time period from which the error message is reviewed.

Field descriptions

Status

The number of the status. The color in which it appears is based on how the status text is configured.

Status text

Designation of the status.

Status type

This selection criteria is used to narrow down the status type shown (depends on the license or project). The following status types, for example, are possible options in the selection criteria:

- Machine status
- Malfunction
- Operation mode
- Operation state
- Program
- ...

Status type designation

Designation of the active status type

Duration

Calculated duration during which the status was active at the machine.

Duration % and quantity

Quantity of the status that occurred and proportion in percent of the total duration.

Shift

Shift number during which the status occurred.

Shift start/ shift end

Shift start and shift end during which the status occurred.

ABC analysis detail application

The ABC analysis detail application provides a presentation of the sum total of all durations that occurred and displays the number of individual postings that are included in it. The data is broken down in three classes A, B, C based on the percentages in relationship to the total duration. This is done in a "Pareto summation", i.e. the individual rows are sorted by their size in descending order into the classes A to C and added to the class until the total sum exceeds the threshold value (to be more precise: threshold value specified 100%). Then the next class will be filled.

ABC-Analyse									
Klassifikation									
Status	Status	Status	Status	Status	Dauer	Anzahl			
Status	Status	Status	Status	Status	Dauer	%	Anzahl	%	
Klassifikation: A									
8	Automatic	100	Operation mode	A	419:48:49	82,54	228	40,71	
					419:48:49	82,54	228	40,71	
Klassifikation: C									
2	Manual	100	Operation mode	C	72:03:48	14,17	161	28,75	
4	Semi-automatic	100	Operation mode	C	16:43:18	3,29	171	30,54	
					88:47:06	17,46	332	59,29	

Individual listing detail application

By selecting a line in the ABC analysis, you can display the separate lines included in the individual listing.

Einzelaufstellung									
Ziehen Sie eine Spalte hierher, um nach dieser zu gruppieren									
Status	Status	Status	Status	Dauer			Schicht		
Status	Status	Status	Status	Start	Ende	Dauer	Schicht	Schichtbeginn	Schichtende
8	Automatic	100	Operation mode	22.07.2010 06:...	22.07.2010 14:...	8:00:00	3	21.07.2010 2:...	22.07.2010 ...
8	Automatic	100	Operation mode	21.07.2010 22:...	22.07.2010 06:...	8:00:00	2	21.07.2010 1:...	21.07.2010 ...
8	Automatic	100	Operation mode	21.07.2010 14:...	21.07.2010 22:...	8:00:00	1	21.07.2010 0:...	21.07.2010 ...
8	Automatic	100	Operation mode	21.07.2010 06:...	21.07.2010 14:...	8:00:00	3	20.07.2010 2:...	21.07.2010 ...
8	Automatic	100	Operation mode	20.07.2010 22:...	21.07.2010 06:...	8:00:00	2	20.07.2010 1:...	20.07.2010 ...
8	Automatic	100	Operation mode	20.07.2010 14:...	20.07.2010 22:...	8:00:00	1	20.07.2010 0:...	20.07.2010 ...
8	Automatic	100	Operation mode	20.07.2010 06:...	20.07.2010 14:...	8:00:00	3	19.07.2010 2:...	20.07.2010 ...
8	Automatic	100	Operation mode	19.07.2010 22:...	20.07.2010 06:...	8:00:00	2	19.07.2010 1:...	19.07.2010 ...
						419:4...			

7 Minor/ Major Stops

Summary

Menu	Operating facilities management → Status analyses → Minor/ major stops
Transaction code	minmaj
Function authorization	minmaj

Usage

This analysis shows the production interruptions (first values posting) for the selected machine. The interruptions are divided into minor and major stops based on their duration. Minor stops are shorter interruptions. They have no noticeable effect on production, because they are absorbed by buffers, for example. However, if the duration of a minor stop exceeds a time threshold defined in advance, it becomes a major stop.

In the list of minor and major stops, all stops with the same cause or with an identical posting text are initially compiled as one item. When a cause is unfolded, these stops are listed individually, whereas the stop that occurred last is shown as the uppermost/ first entry in the list.

Selection criteria

The application provides the following selection criteria:

Workplace

Defines for which workplace the stops are shown.

Date from - to/ shift/ time

Narrows down the time period in which the stop is reviewed.

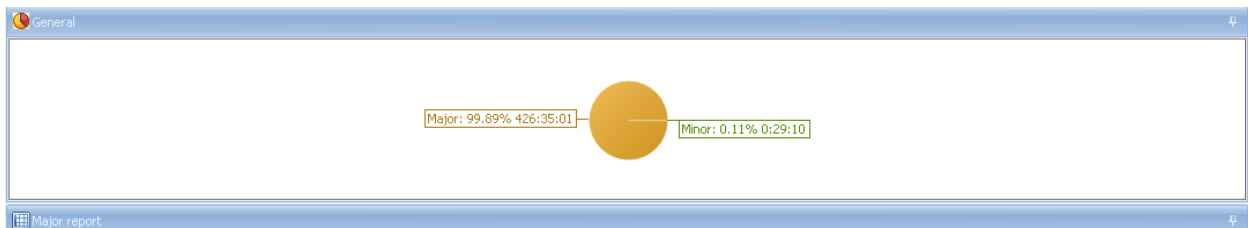
Status type

Narrows down the stop displayed to a status type or to a cause or a posting text.

Minor/ major thresholds

The threshold that is decisive in determining if the stop is considered minor or major. The pre-assignment is set to 5 minutes.

General detail application



Graphic illustration of the total time and how the time is distributed in percent to minor and major stops.

Minor report detail application

The minor report shows the selected stops for the chosen machine with a duration that is shorter than the minor/ major threshold entered.

Minor report							
Drag a column here to group by that column							
Status	Status text	Status type	Status type designation	Duration	Quantity		
Status	Status text	Status type	Status type designation	Duration	%	Quantity	%
99	GENERAL DISTURBANCE	MST	Machine status	0:24:18	0.09	19	33.93
15	MAINTENANCE	MST	Machine status	0:03:38	0.01	1	1.79
22	COLOUR CHANGE	MST	Machine status	0:01:14	0.00	3	5.36
				0:29:10	0.11	23	41.07

Major report detail application

Major report							
Drag a column here to group by that column							
Status	Status text	Status type	Status type designation	Duration	Quantity		
Status	Status text	Status type	Status type designation	Duration	%	Quantity	%
999	FREE SHIFT/BREAK	MST	Machine status	287:43:17	67.37	5	8.93
3	ORDER SHORTAGE	MST	Machine status	105:13:23	24.64	4	7.14
2	SETUP	MST	Machine status	13:58:39	3.27	3	5.36
6	START UP	MST	Machine status	8:05:59	1.90	4	7.14
98	MECHANICAL FAILURE	MST	Machine status	5:26:33	1.27	3	5.36
				426:35:01	99.89	33	58.93

The major report shows the selected stops for the chosen machine with a duration that is longer than the minor/ major threshold entered.

Field descriptions

Status

Number of the status and therefore also the cause for the stop that occurred.

Status text

Status text for the status.

Status type

These selection criteria are used to narrow down the status type shown. The following status types, for example, are possible options in the selection criteria (depending on license and/or project):

- Machine status
- Malfunction
- Operation mode
- Operation state
- Program
- Sequencer A - module
- Sequencer B - programs
- Sequencer B - steps
- Sequencer B - module
- Sequencer B - programs
- Sequencer B - steps
- Sequencer gen. - status

Status type designation

Name of the active status type

Start/ end

Start and end times for the stop

Duration

Calculated duration during which the stop was active at the machine.

Duration % and quantity

Quantity of the status that occurred and proportion in percent of the total duration of the disturbance.

Shift

Shift number during which the stop occurred.

Shift start/ shift end

Shift start and shift end during which the stop occurred.

8 Status Class Report

Summary

Menu	Operating facilities management → Status analyses → Status class report
Transaction code	stclrp
Function authorization	stclrp

Usage

The evaluation provides status information for a specific period of time and a certain number of workplaces. In it, the statuses are consolidated into status classes that were assigned in a previous step. The application provides the user with the ability to run an analysis of the booked status classes.

Selection criteria

The application provides the following selection criteria:

Workplace

Search by machine/ by workplace. You can run a search using wildcards.

Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can run a search using wildcards.

Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can run a search using wildcards.

Company

Search by workplaces/ machines that are assigned to the company that was entered. You can run a search using wildcards.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Short designation

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

Designation

This selection criterion references the name of the machines and workplaces in the machine's master data. At the bottom, only those machines are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders *) in this field.

Date

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, while when selecting by time the selection is based on the start date. Please keep in mind that a selection by shift is only supported for BDE and MDE data, not for WRM data.

Shift(s)/ time

Selection by shift (only BDE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

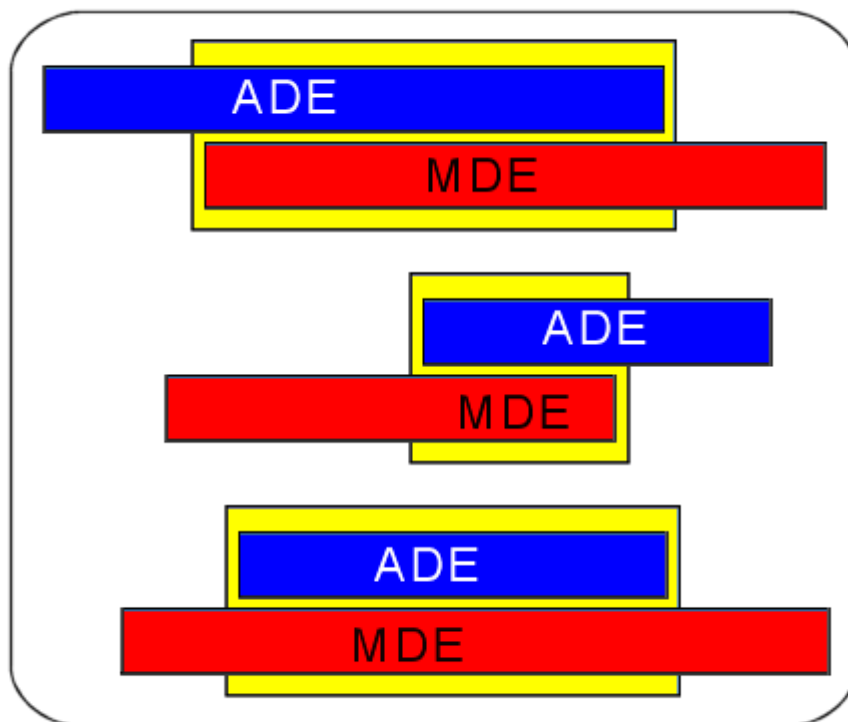
Order/ article

When you choose the option "Order", you must enter an order or an article.

For this kind of evaluation type, only the finished BDE postings are considered. If the order is currently still running on the machine, the time period between the last logon and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only BDE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the BDE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.



The following illustration shows an example of an overlapping of BDE and MDE postings. The BDE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

Additional selection notes

Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Determining a shift-adjusted quantity

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.



Status class report detail application

Tabular presentation of the status class reports

Status class

Status class number

Abbreviation

Abbreviation of the status class

Designation

Description of the status class

Duration

Total time of all statuses within the evaluation period, accumulated to status classes.

%

Proportion of the total duration of the status classes.

Quantity

Total number of all statuses within the evaluation period, accumulated to status classes.

%

Proportion of the number of statuses to the total number of all statuses.

Status class hit list (durations) detail application

Displayed in a bar chart in the status class hit list (durations) detail application are the durations posted to status classes. Displayed here are the status classes that were highlighted in the tabular detail application status class report. The red bars are shown in descending order by duration.

Status class hit list (quantities) detail application

Displayed in a bar chart in the status class hit list (quantities) detail application are the number of status classes. Displayed here are the status classes that were highlighted in the tabular detail application status class report. The red bars are shown in descending order by quantity.

9 Status Class Profile

Summary

Menu	Operating facilities management → Status analyses → Status class profile
Transaction code	stclpf
Function authorization	stclpf

Usage

The status class profile evaluates the status classes by shift, by day, by calendar week or by month over a certain period of time. The result depends on the selection and therefore on the selection criteria made available on the selection panel.

Selection criteria

The application provides the following selection criteria:

Workplace

Search by machine/ by workplace. You can run a search using wildcards.

Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can run a search using wildcards.

Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can run a search using wildcards.

Company

Search by workplaces/ machines that are assigned to the company that was entered. You can run a search using wildcards.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Short designation

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

Designation

This selection criterion references the short name of the machines and workplaces in the machine's master data. The machines or workplaces are displayed that are identical to the string that was entered. You can also run a search using wildcards (placeholders *) in this field.

Date

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, whereas when selecting by time, the selection is based on the start date. Please keep in mind that a selection by shift is only supported for BDE and MDE data, not for WRM data.

Shift(s)/ time

Selection by shift (only BDE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

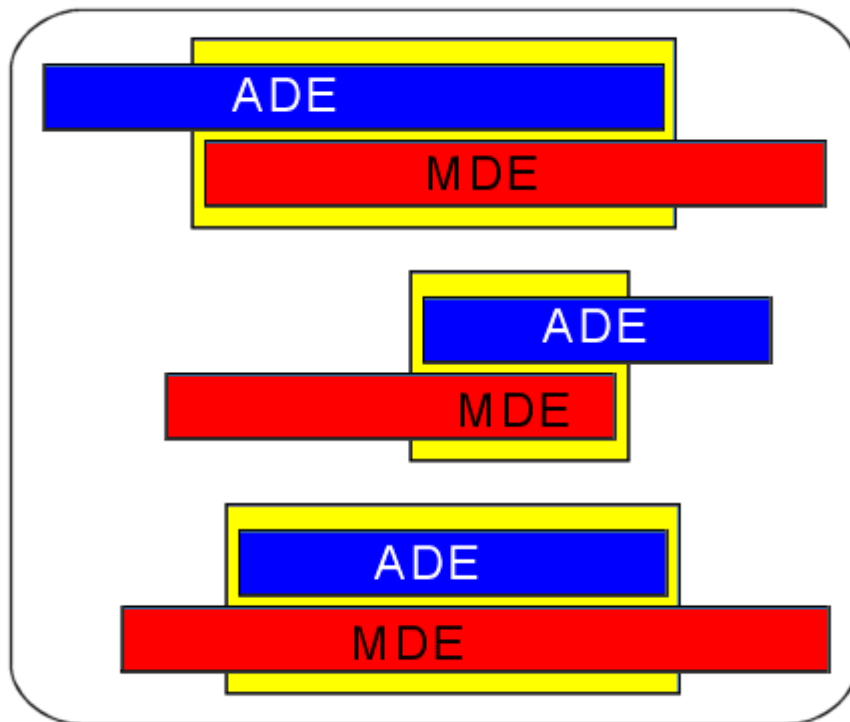
Order/ article

When you choose the option "Order", you must enter an order or an article.

For this kind of evaluation type, only the finished BDE postings are considered. If the order is currently still running on the machine, the time period between the last log on and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only BDE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the BDE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.



The following illustration shows an example of an overlapping of BDE and MDE postings. The BDE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

Additional selection notes

Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Determining a shift-adjusted quantity

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.

Status class profile detail application

Tabular presentation of the status class profile with the following columns:

Shift date/ shift/ calendar week/ year

The durations of the status classes are displayed per shift (i.e. shift date/ shift number). Calendar week, month and year (based on the shift date) are provided as additional sorting/ grouping criteria.

Status class/ abbreviation/ designation

Number, abbreviation and designation used for the status class

Duration

Time duration in which the status class was created/ set within the evaluation period.

Quantity

Number of times that the status class was created/ set within the evaluation period.

Pivot table detail application

You can evaluate status classes based on additional criteria in the pivot table detail view.



The bar colors in the chart are set "arbitrarily" using a color chart defined internally.

10 RPA Report

Summary

Menu	Operating facilities management → Status analyses → RPA report
Transaction code	rpap
Function authorization	rpap

Usage

The RPA report provides status information relating to a workplace/ machine for a specific period of time and a certain number of workplaces. In it, the workplace/ machine related status are consolidated into resource performance accounts that were assigned in a previous step.

Selection criteria

The application provides the following selection criteria:

Workplace

Search by machine/ by workplace. You can also run a search using wildcards.

Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can also run a search using wildcards.

Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can also run a search using wildcards.

Company

Search by workplaces/ machines that are assigned to the company that was entered. You can also run a search using wildcards.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Short name

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. There is an option to use wild cards.

Designation

This field references the designation (in HYDRA: comment) used for the machines and workplaces in the machine master data. Under designation, only those machines are displayed that are identical to the string that was entered. There is also the option to use wild cards (placeholders *) in this field.

Inclusive status for RPA 11

Accounts for the status for RPA 11 (usually, this is the "Production" status).

Date

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, while when selecting by time the selection is based on the start date. Please keep in mind that a selection by shift is only supported for BDE and MDE data, not for WRM data.

Shift(s)/ Time

Selection by shift (only BDE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

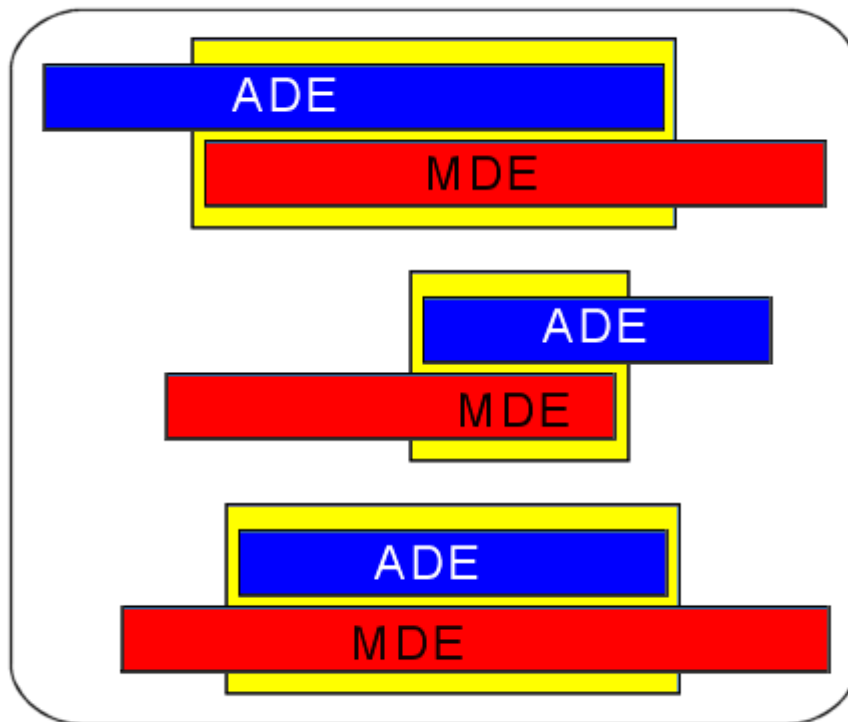
Order/ article

When you choose the option "Order", you must enter an order or an article.

For this kind of evaluation type, only the finished BDE postings are considered. If the order is currently still running on the machine, the time period between the last log on and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only BDE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the BDE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.



The following illustration shows an example of an overlapping of BDE and MDE postings. The BDE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

Additional selection notes

Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Determining a shift-adjusted quantity

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.



RPA report detail application

RPA/ abbreviation/ designation

Number, abbreviation and name used to define the resource performance account.

Duration

Time period during which the status assigned to the RPA was created/ set.

%

Percentage of duration to total duration.

Quantity

Number of times the status assigned to the RPA was created/ set.

%

Percentage of quantity to total quantity.

Duration detail application

In the duration detail application, the resource performance account durations are displayed in a bar chart. The results are displayed by decreasing duration. The color code for each of the resource performance accounts is in accordance with the standard definitions.

Quantity detail application

In the quantity detail application, the number of statuses for the selected machines are displayed in the form of a bar chart, broken down by RPA. The results are displayed by decreasing quantity. The color code for each of the resource performance accounts is in accordance with the standard definitions.

11 PRA Profile

Summary

Menu	Operating facilities management → Status analyses → RPA profile
Transaction code	rpap
Function authorization	rpap

Usage

The RPA profile provides status information for a specific period of time and a certain number of workplaces. In it, the status of resource performance accounts that were assigned in a previous step are consolidated and displayed by shift.

Selection criteria

The application provides the following selection criteria:

Workplace

Search by machine/ by workplace You can also run a search using wildcards.

Group

Search by workplaces/ machines that are assigned to the group that was entered. The selection is made using the field Group in the workplace/ machine configuration. You can also run a search using wildcards.

Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can also run a search using wildcards.

Company

Search by workplaces/ machines that are assigned to the company that was entered. You can also run a search using wildcards.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Short name

This selection criterion references the short name of the machines in the master data. All of the machines or workplaces are displayed that match the string that was entered. You can also run a search using wildcards.

Designation

This field references the designation (in HYDRA: comment) used for the machines and workplaces in the machine master data. At the bottom, only those machines are displayed that are identical to the string that was entered. There is also the option to use wild cards (placeholders *) in this field.

Inclusive status for RPA 11

Accounts for the status for RPA 11 (usually, this is the "Production" status).

Date

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, while when selecting by time the selection is based on the start date. Please keep in mind that a selection by shift is only supported for BDE and MDE data, not for WRM data.

Shift(s)/ time

Selection by shift (only BDE and MDE events) or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

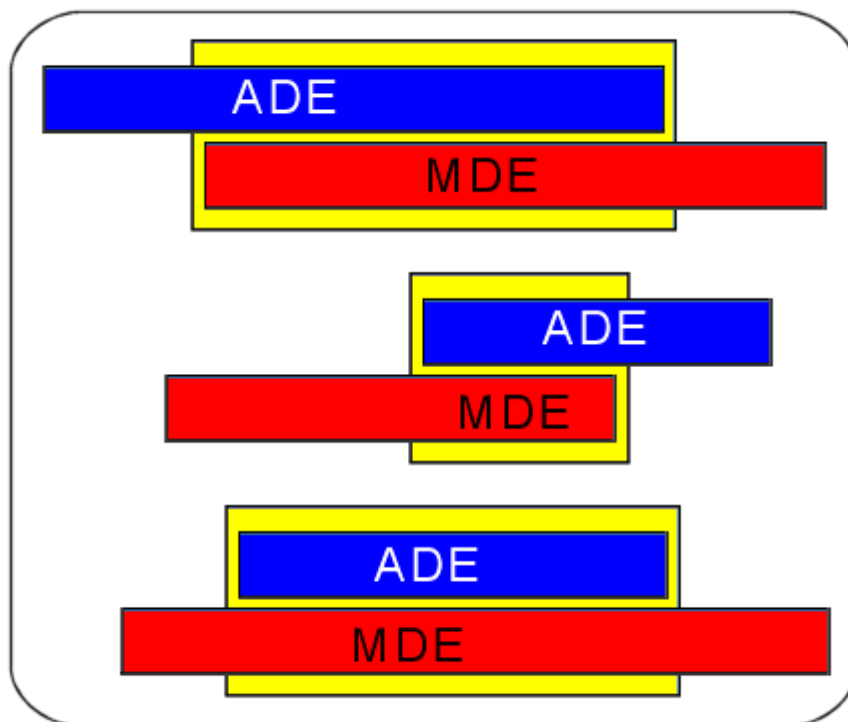
Order/ article

When you choose the option "Order", you must enter an order or an article.

For this kind of evaluation type, only the finished BDE postings are considered. If the order is currently still running on the machine, the time period between the last log on and now is not taken into account. As such, it is by all means possible that there are differences between the machine evaluation and the order-related evaluation. Only BDE postings are taken into account that have started during the evaluation period. If necessary, the selection period must be selected so that the BDE postings that are to be taken into account are within this selection period. For this order-related evaluation, MPDV recommends the shift-related selection option.



The following illustration shows an example of an overlapping of BDE and MDE postings. The BDE postings take priority in this evaluation type. The yellow areas illustrate the results of this evaluation. MDE quantities and durations are calculated on a pro rata basis to achieve the result.



If orders are logged on in parallel at the machine, for this evaluation type, the full machine time (yellow area) and quantity is assigned to each order. The fact that orders are run in parallel will not result in a proportionate calculation.

Resource type/ resource

When choosing the option "Resource", a resource must be entered.

For this kind of evaluation type, here again only the finished postings are considered. For this evaluation type, only the resource postings take priority. The principle is the same as when running an evaluation by order.

Additional selection notes

Long term data

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Determining a shift-adjusted quantity

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.



Overview of detail application

Shift date/ shift/ calendar week/ year

The durations in the resource performance accounts are displayed in groups by shift (i.e. shift date/ shift number).

Calendar week and year are displayed as a possible sorting/ grouping criterion. The calendar week and the year match the shift date.

RPA/ abbreviation/ designation

When delivered from the factory, the durations set for the resource performance accounts 1-10 are listed in the pool of columns. They are stored there with their abbreviation and can be shown as needed.

Duration

Time duration in which the status was created/ set within the evaluation period.

Quantity

Number of times that the status was created/ set within the evaluation period.

Duration detail application

In the duration detail application, the resource performance account durations are displayed in the form of a stacked bar chart. They are displayed in chronological form (X axis) or accumulated to shift date (Y axis). The color code for each of the resource performance accounts is in accordance with the standard definitions.

Quantity detail application

In the quantity detail application, the number of statuses for the selected machines are displayed in the form of a stacked bar chart, broken down by RPA. They are displayed in chronological form (X axis) or accumulated to shift date (Y axis). The color code for each of the resource performance accounts is in accordance with the standard definitions.

12 Status Analysis

Summary

Menu	Operating facilities management → Status analyses → Status analysis
Transaction code	stata
Function authorization	stata

Usage

Because of how flexible it is, the status analysis answers all questions relating to downtimes, malfunction reasons and production times, information that is useful for the shift foreman and team leaders, from the production controller to the production manager.

The recorded production times represent the basic data made available by the status analysis. The status postings are provided here in the finest detail and using the pivot table function, they can be compiled to generate informative reports.

Selection criteria

The application provides the following selection criteria:

Report group

This selection criterion references the report groups. All machines or workplaces are displayed that are assigned to the selected report group.

Cost center

Search by workplaces/ machines that are assigned to the cost center that was entered. You can search using wild cards.

Workplace

Search by machine/ by workplace. You can search using wild cards.

Responsibility area

This selection criterion references the responsibility area in the machine master data. Keep in mind that only machines are displayed that the user has also assigned responsibility areas to.

Company

Search by workplaces/ machines that are assigned to the company that was entered. You can search using wild cards.

Group

Search by workplaces/ machines that are assigned to the group that was entered.

Date

The period of time from which data should be selected.

When selecting via shift(s), the shift date is evaluated, whereas when selecting by time the selection is based on the start date. Please keep in mind that a selection by shift is only supported for ADE and MDE data, not for WRM data.

Shift/ time

Selection by shift or by time period. If no shift has been selected, all shifts are considered.

The two times each refer to the start or to the end of the date periods listed above.

Additional selection notes**Long term data**

If the selection period exceeds the period for the online data area, the system applies the implicit solution and selects the medium-term data area as well. Therefore, there is no need for an explicit activation in order to be able to access the medium-term data set.

Determining a shift-adjusted quantity

This option known from MDE 7.2 is set by default in MOC. What this means is that postings that were generated as a result of a shift change are not considered when determining the quantity. All that is accounted for here is the exact moment when a machine status was set. If this moment is outside of the evaluation interval, in this case the output will be 0.

Status analysis detail application

The results that were found are displayed in this detail application in tabular form. The results answer the question: When was a certain status recorded for which machine and for how long?

The following columns are shown, among others:

Workplace/ short designation

Number and short name of the workplace

Beginning/ end

Beginning period or ending period of the status respectively.

Shift date/ shift number

Shift date and shift in which the status applied.

Status/ status text

Number and designation of the active status

Duration

Duration of the status

Quantity

The Quantity field reflects how often a status applied. Because only one status is displayed here at a time, generally a 1 is displayed as the quantity.

A 0 is displayed if the status was set due to an automatic status change that took place during a shift change and it is no different than the previous status (before the shift change).

Pivot table detail application

Available in the detail application "Pivot table" are data found that are used for a pivot analysis. Functions such as those known from Microsoft Excel® can be used for this purpose.



The colors used in the graphic do not depend on the colors used for status configuration.